

Due in class tuesday February 7th

This is the first problem set. Please present clean drawings/sketches wherever relevant, and show all calculations up to reasonable detail.

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1. (a) [**1+1 pts.**] Look up and report the values of the *permittivity of free space*, ϵ_0 , and the *permeability of free space*, μ_0 , in SI units. Include the units of each quantity.

(b) [**2 pts.**] Look up and report Coulomb's law in SI units and in CGS units.
2. We will ignore the z direction for this problem. Consider the following points on the x - y plane: the point A with coordinates $(0, a)$, the point B with coordinates $(0, -a)$, and the point P with coordinates $(r, 0)$.

(a) [**7 pts.**] A particle at point P feels two forces, both of equal magnitude F_0 . One force points in the direction from A towards P (along the direction of the line AP), and the other acts towards B (along the direction of the line PB). Find the total force acting on the particle. Note: Finding a vector means finding both its magnitude and its direction!!

(b) [**7 pts.**] Imagine two vectors, each having the same magnitude E_0 , acting on point P . One points from P towards A (along the direction of the line PA), and the other points from P towards B (along the direction of the line PB). Find the result of adding the two vectors.
3. [**2 pts.**] Look up and report what a *vector field* is.
4. [**7 pts.**] The electric field in some region is given by

$$\mathbf{E} = 5xy\hat{i} + 2z^2\hat{j}$$

where $\hat{i}$ and $\hat{j}$ are unit vectors in the x and y directions. Find the force experienced by a charge q that is placed at the point $(x, y, z) = (1, -1, 1)$.

5. **[5+5 pts.]** Look up or recall Stokes' theorem and Gauss's divergence theorem, and report each clearly.
6. Here L and d are positive charges, with $d > L$. Also, Q and q are positive charges.
- (a) **[6 pts.]** Five point charges are placed on the x -axis, at the points with x -coordinates $L/5$, $2L/5$, $3L/5$, $4L/5$ and L . Each of these carry the charge Q/L .
A point charge q is placed, also on the x -axis, at the point $(d, 0, 0)$.
Find the total force experienced by the charge q .
- (b) **[7 pts.]** N point objects, each carrying charge Q/N , are placed on the x -axis, at the points with x -coordinates

$$L/N, 2L/N, 3L/N, \dots, L$$

Find the force experienced by a charge q , placed at the point $(d, 0, 0)$.
Comment: When N is very large, we can regard this as a rod of continuous charge extending from the origin to $(L, 0, 0)$, carrying total charge Q . This semester we will learn to calculate the forces/fields due to continuous charge distributions of various shapes.